

Ordinals and Typed Lambda Calculus

Defining Ordinals in Martin-Löf Type Theory

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Hilbert Ordinals

Definition

Hilbert ([1925] 1967) (originally published as (Hilbert 1926)) axiomatically defined the ordinals of the (cumulative) second number class. Let $\mathbf{N}$ and $\mathbf{O}$ be unary predicates representing natural and ordinal numbers, respectively.

$\mathbf{N}$ zero,

$\forall n [\mathbf{N}n \rightarrow \mathbf{N}(\text{succ}(n))],$

$\{P \text{ zero} \wedge \forall n [Pn \rightarrow P(\text{succ}(n))]\} \rightarrow \forall n (\mathbf{N}n \rightarrow Pn),$

$\mathbf{O}$ zero,

$\forall n [\mathbf{O}n \rightarrow \mathbf{O}(\text{succ}(n))],$

$\forall n [\mathbf{N}n \rightarrow \mathbf{O}(f(n))] \rightarrow \forall n [\mathbf{O}(\lim f(n))],$

$\{P \text{ zero} \wedge \forall n [Pn \rightarrow P(\text{succ}(n))] \wedge \forall n [P(f(n)) \rightarrow P(\lim f(n))]\} \rightarrow \forall n (\mathbf{O}n \rightarrow Pn).$

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Discussion

Can we define a data type for representing ordinal numbers from Hilbert's axioms?

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Remark

Cumulative second number class have been studied by various authors (see, e.g., (Church and Kleene 1937; Coquand, Hancock and Setzer 1997; Howard 1972; Martin-Löf 1984, [1970] 2001; Stenlund 1972)).

Martin-Löf's Ordinals

Remark

Martin-Löf (1984) defined ordinals in his type theory. He did not mention Hilbert but Cantor when given his definition.

Introduction rules

The introduction rules for Martin-Löf's ordinals are the following ones.

$$\frac{}{\text{zero} : \text{Nat}}$$

$$\frac{n : \text{Nat}}{\text{succ } n : \text{Nat}}$$

$$\frac{}{\text{zero}_o : \text{On}}$$

$$\frac{n : \text{On}}{\text{succ}_o n : \text{On}}$$

$$\frac{f : \text{Nat} \rightarrow \text{On}}{\text{lim}_o f : \text{On}}$$

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